

RCTs against the Machine: Can Machine Learning Prediction Methods Recover Experimental Treatment Effects?

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Abstract: We investigate how successfully machine-learning (ML) prediction algorithms can be used to estimate causal treatment effects in electricity demand applications with nonexperimental data. We use three prediction algorithms—XGBoost, random forests, and LASSO—to generate counterfactuals using observational data. Using those counterfactuals, we estimate nonexperimental treatment effects and compare them to experimental treatment effects from a randomized experiment for electricity customers who faced critical-peak pricing and information treatments. Our results show that nonexperimental treatment effects based on each algorithm replicate the true treatment effects, even when only using data from treated households. Additionally, when using both treatment households and nonexperimental comparison households, standard two-way fixed effects regressions replicate the experimental benchmark, suggesting little benefit from ML approaches over standard program evaluation methods in that setting.

JEL Codes: C55, L94, Q41

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IN THE LAST THREE DECADES, a central focus of applied economics research has been estimating causal effects of policies, market rules, or other factors on market outcomes, including consumer behavior. Well-designed and well-executed randomized

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controlled trials (RCTs) offer a high degree of internal validity within the randomized population or sample for understanding the causal effects of such interventions and have been applied in many settings, including development economics (e.g., Banerjee and Duflo 2009), education (e.g., Krueger 1999), labor economics (e.g., List and Rasul 2011; Jones et al. 2019), and environmental and energy economics (e.g., Ferraro and Price 2013; Allcott and Rogers 2014; Jessoe and Rapson 2014). However, RCTs are costly, can take substantial amounts of time, and may not be feasible in certain settings due to regulations prohibiting the requisite randomization of subjects into treatment and control groups or firm or government unwillingness to implement experimental designs. In particular, these challenges can hamper efforts to implement RCTs designed to explore the impacts of pricing and other interventions on electricity demand, which is the focus of this study. More generally, researchers seeking to estimate causal treatment effects may only have access to nonexperimental data.

In such settings lacking randomized treatment, machine-learning (ML) algorithms paired with program evaluation tools offer a potential alternative approach to estimate treatment effects. In data-rich settings, ML algorithms can be used to explain variation in a variable of interest by using regularization techniques to select the most relevant covariates for making out-of-sample predictions. In both data-rich and non-data-rich settings ML approaches can also allow for more flexible functions that can capture complex interactions and nonlinearities in covariates. These prediction methods can, in turn, be used as a credible counterfactual that, when compared to observed data on particular outcomes, can reveal the effect of a policy change. These methods are particularly valuable in settings where outcomes are cyclical or seasonal and detailed, high-frequency data are readily accessible by researchers (Burlig et al. 2020; Christensen et al. 2021). Electricity demand is one such setting that provides the requisite data and, given the importance of aligning energy use with marginal private and social costs that can vary considerably throughout the day and across seasons, understanding the causal implications of strategies to shape electricity demand is of immediate policy importance. These issues will become even more salient as the electricity sector moves toward greater reliance on intermittent sources of power like wind and solar.

In this study, we compare treatment effect estimates from a randomized electricity pricing and information experiment to nonexperimental effects estimated using both (1) standard methods that apply two-way fixed effects (TWFE) regressions to the observed outcome of interest (here, electricity consumption) and (2) using three commonly used ML counterfactual prediction methods (XGBoost, random forests, and least absolute shrinkage and selection operator [LASSO]) in a “prediction-error” framework, where predicted outcomes serve as counterfactuals. We partnered with the

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nonprofit Pecan Street Inc. to access high-frequency energy-use data matched with household characteristics for a set of households in Austin, Texas, that were part of this experiment (also analyzed by Burkhardt et al. [2019], henceforth Burkhardt et al.) and similar households that were not subject to the experiment. We train our ML models using pretreatment hourly consumption data, household characteristics, weather, and interactions among these variables to make out-of-sample predictions for electricity consumption during the treatment period that we use as household-by-hour-specific counterfactual consumption. We then compare these counterfactuals to the observed consumption in a fixed-effects regression framework to estimate nonexperimental treatment effects. Specifically, having computed prediction errors—that is, the difference between observed household-by-hour consumption and its predicted counterfactual using historical data—we regress that error on treatment indicators in a standard two-way fixed effects framework. The intuition behind this approach is if, conditional on fixed effects, treated households exhibit lower electricity consumption beyond what would otherwise be predicted by an ML model—that is, a negative prediction error—and that this negative prediction error is more pronounced during treated time periods, then this provides evidence for a negative treatment effect. We implement this approach for four treatments, including ones with both true significant effects and null effects according to the experimental benchmark, to assess the ability to replicate both kinds of results.

To assess the value of the ML counterfactual prediction methods in electricity demand contexts, we implement our procedure using three different samples in which we pair the treatment groups' data with (a) the original experimental control group data, (b) a nonexperimental comparison group comprising similar households that were not part of the randomized experiment, and (c) no additional comparison group—that is, with only data on treated households. The first sample forms our experimental benchmark using both observed and predicted data. The second sample forms a nonexperimental replication sample, which we use to estimate treatment effects. The third sample follows the same format but uses data from the treatment group only, forgoing the nonexperimental comparison group entirely, with predicted consumption serving as a counterfactual. In each case, we calculate treatment effect estimates both in a conventional TWFE framework applied to observed electricity use data as the dependent variable, and also in a prediction-error framework in which the TWFE approach is applied to the prediction error of electricity use, where ML-predicted electricity use serves as counterfactual consumption. This multisample approach demonstrates how and when ML-based counterfactual prediction does or does not add value over conventional TWFE approaches.¹

1. Our approach corresponds with ad hoc approaches in which applied researchers use machine-learning algorithms to generate counterfactuals, which are then used to estimate standard fixed-effects specifications (see, e.g., Liu et al. 2021). Econometricians have developed more

We uncover several important results. First, each ML counterfactual prediction approach replicates the RCT's true treatment effects (both true significant and null effects), even with data on treated households only. This result suggests that ML counterfactual prediction methods can construct a valid counterfactual when researchers do not have a credible set of comparison units. Second, when using treatment households along with nonexperimental comparison households, we find that simpler TWFE methods can replicate the experimental benchmark, while ML-predicted counterfactuals offer little benefit relative to traditional program evaluation approaches with the nonexperimental comparison group. Third, we find little difference in the ability of each ML algorithm to replicate the experimental benchmark, despite nontrivial differences in predictive accuracy across algorithms. This last result suggests that predictive accuracy is an imperfect proxy for how well ML prediction methods will perform in estimating causal treatment effects. Additionally, we explore what combination of variables (e.g., weather, household characteristics, and covariates representing diurnal, weekly, and seasonal patterns) are necessary for replication, which provides actionable insight for applied researchers. We find that ML counterfactual prediction methods can replicate experimental treatment effects remarkably well with relatively little data on predictors.

These methods and results may be particularly useful in analyzing energy demand and how pricing and other policy interventions intended to change consumer demand choices affect actual consumption behavior. The growing penetration of smart electricity meters has increased the temporal granularity of consumption data, which both presents researchers with potentially valuable opportunities for better policy evaluation (Ghanem and Smith 2021) and makes it increasingly possible to implement time-varying prices that could increase the efficiency of electricity market outcomes. In practice, most policy evaluation is performed with nonexperimental data, so new ML-based methods applied to newly available rich data sets offer an implementable opportunity for improving the estimation of causal effects.

1. RELATED LITERATURE

We apply ML counterfactual prediction methods to study policies intended to reduce electricity consumption at times of peak demand. Specifically, we consider when such ML methods can replicate the results of an electricity pricing experiment that was evaluated by Burkhardt et al. Economists have long been advocates for electricity prices

sophisticated double-debiased machine-learning algorithms to incorporate the first-step prediction within program evaluation methods to estimate causal parameters and, importantly, account for error that arises in the first-step prediction (e.g., Chernozhukov, Chetverikov, et al. 2018; Chang 2020). We do not extend our analysis to this class of estimators, and these estimators may or may not present gains compared to standard methods. We leave this analysis for future research.

that reflect temporal variations in the costs of electricity production to encourage consumers to shift their use away from high-cost hours (Borenstein and Holland 2005; Harding and Sexton 2017; Borenstein and Bushnell 2019). Real-time pricing would produce the most substantial efficiency gains relative to time-invariant pricing (Borenstein 2005), but it can face significant political hurdles (Borenstein 2007; Joskow and Wolfram 2012) and is rarely implemented. Other approaches, such as critical-peak pricing, can approximate real-time pricing and yield much of its welfare gains (Blonz 2022). Consumers will respond to various forms of time-varying prices (Wolak 2011); however, in many cases, consumer responses have been quite small (Jessee and Rapson 2015; Fabra et al. 2021), which could be rational if attention costs are large (Sallee 2014). Consumer responsiveness is also often heterogeneous in a policy's effects (Fowlie et al. 2020; Prest 2020), and information can help increase price responsiveness (Jessee and Rapson 2014), as can automation (Harding and Lamarche 2016; Gillan 2018; Blonz et al. 2021). We contribute to this literature by demonstrating how and when ML counterfactual prediction methods can be useful in evaluating energy demand policies.

Our analysis also speaks to a deep literature spanning the last 35 years that has evaluated how well common program evaluation methods using observational data perform in recovering experimental treatment effects. This work has provided general insight into the usefulness of different empirical approaches in different settings (Cook et al. 2008; Imbens and Wooldridge 2009). These “design-replication studies”—sometimes referred to as “within-study designs”—focused initially on replicating the results from an experimental job-training program, starting with LaLonde (1986). In these settings, researchers attempt to replicate the results of an experimental benchmark, typically generated from a randomized field experiment, without using the experimental control group. To do so, authors gather a nonexperimental comparison group to serve as a credible replacement for the experimental control group and then apply common research designs, including TWFE, propensity-score matching, and covariate matching, as well as regression discontinuity and instrumental variables in certain settings.²

In recent years, the advent of high-frequency and feature-rich data sets has spurred a rapid development and adoption of ML tools to predict counterfactuals that can be

2. The initial conclusions from this line of inquiry suggested that early propensity-score matching designs failed to replicate randomized results, although this conclusion has been subject to lively debate, finding cases in which the method does succeed, particularly among subsamples where treatment and comparison groups are demonstrably similar (e.g., Heckman et al. 1998; Dehejia and Wahba 1999, 2002; Dehejia 2005; Smith and Todd 2005). More recently, design-replication studies that pair covariate matching with panel designs have fared comparatively better than early results that focused on propensity-score matching, likely due to the ability to control for both observable and unobservable heterogeneity by design (e.g., St. Clair et al. 2014; Ferraro and Miranda 2017). However, some have found that nonexperimental results from matching and panel data fail to replicate even when exhibiting strong balance and common pretrends (Wichman and Ferraro 2017).

used for the estimation of causal effects. While the idea of estimating causal effects in reference to an unobserved counterfactual is not new (Rubin 1974), recent advances in ML algorithms and increased availability of rich data sets have greatly improved economic researchers' prediction accuracy (Varian 2014, 2016). Although counterfactual prediction using ML has the flavor of synthetic control approaches (e.g., Abadie and Gardeazabal 2003; Abadie et al. 2010), ML approaches permit much more flexible nonlinear, high-dimensional models that can deliver superior predictions. More generally, the economic literature has recently seen an explosion of research incorporating ML into econometric approaches, with a variety of applications in various contexts, including prediction problems, high dimensionality, and heterogeneous treatment effects (e.g., Athey and Imbens 2016; Athey et al. 2017; Belloni et al. 2017; Chernozhukov, Chetverikov, et al. 2018; Wager and Athey 2018; Chang 2020). The theoretical frontier of this research for program evaluation combines machine-learning and difference-in-difference estimators in a double/debiased machine-learning framework (e.g., as in Chang 2020). Our approach is more ad hoc, in that we use machine learning solely to generate a counterfactual prediction, which we then incorporate into standard program evaluation methods.

Among ML-based counterfactual prediction approaches, specific ML algorithms that have recently been used to predict counterfactuals include LASSO and random forests (e.g., Liberman et al. 2018; Abrell et al. 2019; Burlig et al. 2020; Dueñas et al. 2021), XGBoost (e.g., Souza 2019; Christensen et al. 2021; Dueñas et al. 2021), matrix completion (Athey et al. 2021), and deep neural networks (Hartford et al. 2016). The application in this study is energy, but these methods are used in many areas in the literature just referenced, ranging from studying consumer credit in developing countries to the effects of COVID-19 on international trade. In contrast to other recent applications of ML tools in the context of causal inference, such as to estimate heterogeneous treatment effects (e.g., Athey and Imbens 2016; Knittel and Stolper 2019; Prest 2020; Cunningham et al. 2021) or firm unobservables (e.g., Davis et al. 2021), the focus of this study is counterfactual prediction for ex post policy evaluation. These approaches are promising, particularly in settings in which an observational, nonexperimental comparison group cannot be found. However, there have been relatively few formal comparisons of these ML counterfactual prediction tools to a randomized benchmark.³

2. EMPIRICAL SETTING AND DATA

We use data from a randomized experiment conducted by Pecan Street Inc., a non-profit research institute based in Austin, Texas. Pecan Street collects detailed use and generation (for rooftop solar) data to enable analysis of various aspects of electricity, ranging from the effects of pricing programs to smart home technology and electric vehicles.

3. Two notable exceptions include Chernozhukov, Chetverikov, et al. (2018) and Chernozhukov, Demirer, et al. (2018).

Pecan Street monitors energy use in more than 1,000 homes at high temporal resolution (down to the minute level). Throughout 2013 and 2014, Pecan Street conducted a randomized controlled trial (RCT) on about 250 households in Austin's Mueller neighborhood to assess the effect of four alternative pricing and information treatments on electricity use during peak demand hours on particularly hot summer days.

2.1. Benchmark Experimental Estimates

The Pecan Street RCT that we use as a benchmark was evaluated by Burkhardt et al. Their approach offers "ground-truth" estimates of a set of treatment effects to which we can compare and thereby evaluate our ML counterfactual prediction approaches.

Pecan Street recruited households into the experiment by offering a \$200 participation incentive; 280 households ultimately participated, with the majority in the Mueller neighborhood in Austin. The goal was to test whether (and what type of) incentives and information about critical-peak demand events would generate short-run conservation behavior during the peak period (4 p.m.–7 p.m.). The four treatment arms were as follows:

- *Passive information* ("Portal"): Households were given access to an online portal with information about their energy use.⁴
- *Active information* ("Text"): Households were sent text messages the day before each critical-peak event, which stated: "A Pecan Street Project critical peak event is taking place tomorrow from 4 PM to 7 PM."
- *Active information + recommendation* ("Text+Rec."): Households received the same text message as the Text treatment arm, with the addition of one of the following recommended actions: "Pre-cool your home," "Reduce your air conditioning usage," or "Do not use your clothes dryer."
- *Pricing* ("Pricing"): Households received a text message the day before each pricing event, which stated: "Tomorrow is a Critical Peak Pricing event. Your experimental electric rate will be \$0.64 per kilowatt hour from 4 PM–7 PM. Pecan Street Inc. Pricing."

Customers experienced 27 critical-peak pricing events between June 2013 and September 2014.

Burkhardt et al. evaluated the experiment using minute-level electricity consumption data at the household level. To estimate treatment effects, Burkhardt et al. regressed electricity consumption on a dummy variable, T_{ijt} , which indicates that household i is in treatment group j and received the treatment during "event" time t (which

4. All treatment arms had access to this passive information; households in the control group did not.

captures whether a critical-peak pricing event took place on a given day in the post-treatment period). In addition to the time-varying treatment indicator, Burkhardt et al. included household and quarter-hour-of-sample fixed effects as well as residual time and treatment interactions that are standard in TWFE-style difference-in-differences regressions. Standard errors are clustered at the household level. As a result, Burkhardt et al.'s TWFE specification produces a "triple-difference" estimate of an average treatment effect that can be interpreted as the change in electricity use across (i) treatment-control households, (ii) event and nonevent days, and (iii) critical-peak and non-critical-peak hours. Burkhardt et al. find that the Pricing treatment reduced hourly electricity use by 0.39 kilowatt-hour (kWh) ($\sim 14\%$) per hour, although the Portal, Text, or Text+Rec. treatments were all estimated to have no economically or statistically significant impact on consumption.

We acquired the experimental data through Pecan Street independently in 2020. Between the time that Burkhardt et al. acquired the data and when we did, Pecan Street implemented small changes to the historical data to improve data quality; as a result, our data (and, hence, our benchmark results) differ slightly from those presented in Burkhardt et al.'s analysis. The major difference is that our data set features 249 households in the RCT (vs. 256 in Burkhardt et al.) because Pecan Street found some data problems for a small number of homes. Although our sample differs slightly from Burkhardt et al.'s, the resulting treatment effects are nearly identical.

Additionally, we extend their analysis by drawing upon more historical data to train our ML models and we also consider an additional, nonexperimental sample of comparison households that were monitored by Pecan Street during the same time frame but were not part of the experiment, thereby creating a larger sample. The final difference is that for computational reasons, in training our ML models, we aggregate minute-level data to an hourly time step. Testing shows that the minute-level data produce the same treatment effect estimates as the hourly-level aggregation, which is intuitive because the treatment was imposed over three-hour windows. With data aggregated to the hourly level, we cannot use the quarter-hour-of-sample fixed effects in Burkhardt et al.'s TWFE specification, so we use hour-of-sample fixed effects instead.

With slight changes in our sample relative to the Burkhardt et al. benchmark, our empirical approach is adjusted accordingly. We aggregate electricity use data to the household-by-hour level but retain the central triple-difference approach. As a result, the finest level of time fixed effects we use is hour of sample (which is discussed in detail in sec. 4). Our results, however, exploit the same variation across households, event days, and event hours.

2.2. Data

In addition to the high-resolution household electricity consumption data described earlier, the data set includes household characteristics: square footage, house age, whether solar photovoltaic (PV) is installed and how much, and whether the house

is a single-family residence. Finally, it includes local weather variables at the hourly resolution: temperature, humidity, dew point, visibility, apparent temperature, pressure, and precipitation intensity. We use all of these characteristics in our ML predictive models.

Because we identify treatment effects using a TWFE strategy, we assume parallel trends in energy consumption across the treatment and control groups. We visually assess the validity of this assumption in figure 1 by graphing average weekly consumption for each group both before and after the treatment period begins. Because the treatments are primarily focused on changing behavior during peak hours, we show not only average consumption across all hours of the day (top panel) but also average consumption only during peak hours (4 p.m.–7 p.m.) (bottom panel).

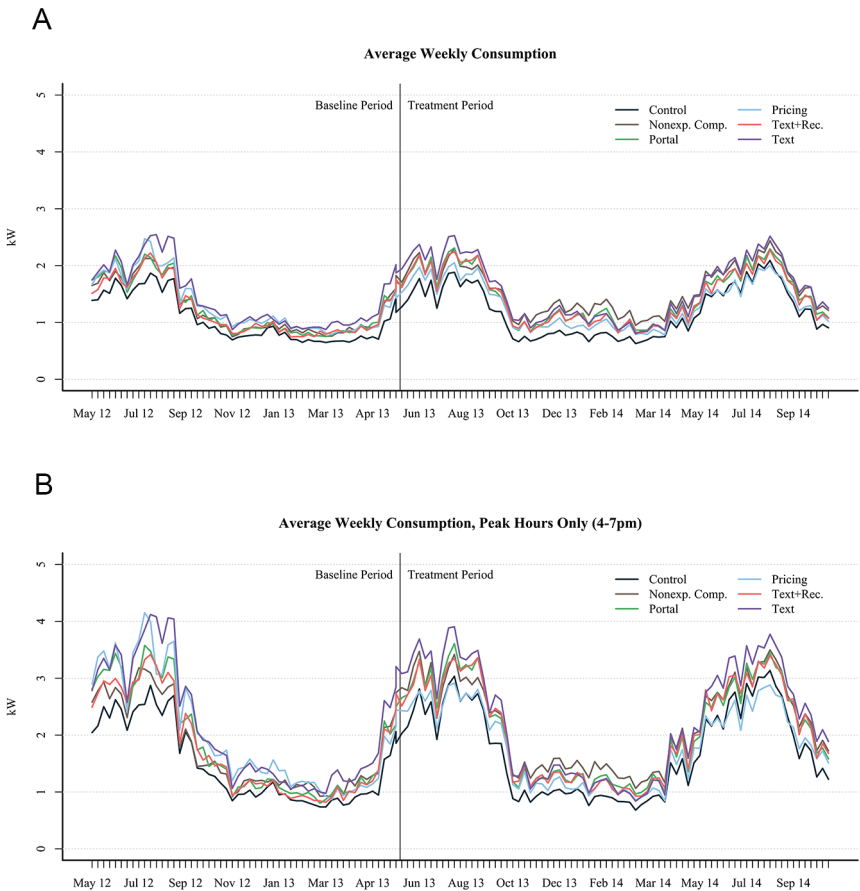


Figure 1. Weekly average energy consumption for all hours (A) and peak hours (B), by group

The seasonal pattern appears similar across control groups, suggesting that the parallel trends assumption is reasonable. However, the treatment groups appear to have somewhat higher baseline consumption (and, indeed, we find these differences to be statistically significant). To the extent that these differences are time invariant, they will be absorbed by household fixed effects in the TWFE specification. Further, even time-varying differences will be absorbed to some degree by the estimator, which as discussed in section 3.3.1 effectively amounts to a triple-difference setup (which compares event days to nonevent days in the treatment period).

3. METHODS

In this section, we describe our empirical approach in detail. We first describe how we implement our counterfactual predictions via three ML algorithms that are used to generate a “prediction error” that reflects the difference between the ML counterfactual and our observed data. We then show how we use this prediction error in a two-way fixed effects (TWFE) regression framework to estimate nonexperimental treatment effects using different samples of comparison households. Last, we discuss the criteria used to compare nonexperimental treatment effects with our randomized benchmark.

We train three commonly used ML models (XGBoost, random forest, and LASSO) on a year of historical hourly, household-level electricity consumption data. The training period is June 2012 through May 2013, which are the 12 months immediately preceding the rollout of the RCT. Each model is used to generate out-of-sample household-by-hour predictions during the experimental period, June 2013 through October 2014. These predictions serve as a counterfactual prediction for what a household would have consumed.

Each ML approach uses unit-level household characteristics and hourly weather data to inform our prediction models. The household characteristics that we include are square footage, house age, whether solar PV is installed and how much, and whether the house is a single-family residence. Weather data include temperature, humidity, apparent temperature (i.e., the “feels like” temperature), dew point, visibility, pressure, and precipitation intensity. We include squared terms of all weather variables except visibility. We also capture predictable diurnal and seasonal consumption patterns by including flexible time covariates: hour of day, day of week, week of year, month of year, weekend, and peak hour. We include interactions of all these covariates, either implicitly (in random forests and XGBoost, which by their nature construct flexible interactions) or explicitly (in LASSO). This allows for complex, highly nonlinear relationships between the variables.

3.1. ML Training

ML models simultaneously allow for flexible, highly nonlinear relationships and avoid overfitting through cross validation (CV). Conceptually, CV splits a data set into

“training” and “test” sets; each model is estimated on the training set and then used to generate predictions on the test set. One can compare the out-of-sample predictive accuracy of alternative models by evaluating which perform best on the test set, such as in terms of root mean square error (RMSE). Models with few covariates tend to lack sufficient information to generate good predictions, whereas overly complex models tend to overfit training data and produce poor predictions out of sample. CV allows one to find the model (or models) that strike the right balance by evaluating the out-of-sample predictive accuracy for a series of increasingly complex models. This process is called “tuning.”

We tune each model using K -fold CV, where each observation is assigned to one of K “test” folds that are iteratively held out for evaluation. We use $K = 10$ for LASSO and random forests and $K = 5$ for XGBoost due to computational constraints. As in Burlig et al. (2020), we assign observations to folds in weekly blocks to ensure that the training data incorporate the observed temporal autocorrelation in energy consumption.

To avoid potential confusion, we pause to be explicit about our CV and counterfactual prediction periods. Our goal is to generate counterfactual predictions for the RCT’s treatment period, June 2013 through October 2014. We call this the “prediction period,” to distinguish it from the test set. The test sets are based entirely on pretreatment data during the preceding year, which we refer to as the “training period”: June 2012 to May 2013. In K -fold CV, the training period is divided into training and test sets, but those test sets consist entirely of data during the “training period.” This approach correctly ensures that the counterfactual predictions during the prediction period are not improperly derived using prediction period data.

Our predictive models are “pooled,” meaning that each method uses the same model across households. We also considered estimating household-specific predictive models, but those models tended to achieve worse out-of-sample predictive accuracy than the pooled models. This finding suggests that household-specific models can overfit by failing to incorporate cross-household information. While our “pooled” models are the same across households (for a given ML approach and sample), the predictions vary by household because of the use of household characteristics and their interactions with other variables.⁵

We now describe the details of each ML approach: XGBoost, random forest, and LASSO. In each case, we tended to favor default, “off-the-shelf” model options in the relevant statistical packages in R because these are the methods that many practitioners would likely use. We tune hyperparameters for each model as described in the subsequent sections; for hyperparameters or options not explicitly discussed, we use the default options in the relevant R package.

5. In a sensitivity analysis discussed in sec. 5, we use household fixed effects instead of household characteristics when training the ML model, which allows for some household specificity in the prediction models because the fixed effects can interact with other covariates.

3.1.1. *XGBoost*

XGBoost, also referred to as a “gradient boosted machine” (GBM) or boosted trees, uses regression trees to generate predictions. Regression trees use recursive partitioning of a data set along each covariate (e.g., is the temperature above or below 80 degrees?). This recursive partitioning creates rectangular regions of the covariate space, denoted $R_1, \dots, R_J$, such that each observation falls in precisely one region R_j . For a given tree, the predicted value of the dependent variable in each region equals the sample average of observations in that partition, denoted $\hat{Y}_{R_j}$. The data set is split recursively many times to automatically form regions representing flexible interactions between covariates (e.g., generating a distinct prediction for energy use when the temperature is above 80 degrees on a Sunday at 2 p.m. in a house larger than 1,500 square feet). In a standard regression tree, the goal of the algorithm is to find the sequential splits on covariates to minimize a loss function given by the sum of squared prediction errors:

$$\sum_{j=1}^J \sum_{i \in R_j} (Y_i - \hat{Y}_{R_j})^2. \quad (1)$$

Under standard regression trees, very large, complex trees are grown with many regions R_j that may fit training data well but nonetheless perform poorly out of sample due to overfitting. Therefore, standard trees are pruned back by eliminating splits along covariates (leading to fewer regions R_j). The exact splits to be “pruned” are then chosen by cross-validation. In such cross-validation, an out-of-sample RMSE is calculated for each tree of varying complexity, and typically the tree yielding the lowest RMSE is chosen. For more details, see chapter 8 of James et al. (2013).

XGBoost is an advancement upon simple regression trees. XGBoost grows many regression trees sequentially that each aim to predict the residuals not yet explained by the previous trees. Specifically, boosting starts by estimating an initial regression tree, $\hat{f}^1(x_i)$, for the outcome, y_i , then shrinks it by the learning rate denoted $\eta \in (0, 1)$. We use $\eta = 0.3$, the default in the widely used XGBoost package. The resulting residual is calculated as $\epsilon_i^1 = y_i - \eta \hat{f}^1(x_i)$. Next, a new regression tree $\hat{f}^2(x_i)$ is estimated on the residual, ϵ_i^1 , instead of the outcome y_i , to explain some of the remaining variation, generating a new residual, and so on. This process is repeated T times, where on iteration b the residual is updated as $\epsilon_i^b = \epsilon_i^{b-1} - \eta \hat{f}^b(x_i)$. The final predictive model is the aggregation of the T trees $\hat{f}(x_i) = \sum_{b=1}^T \eta \hat{f}^b(x_i)$. Chapter 8 of James et al. (2013) presents an intuitive introduction to boosting, and Chen and Guestrin (2016) presents a detailed description of the implementation we use, XGBoost in R.

We tune XGBoost’s hyperparameters as shown in table 1, leaving the remaining ones at their default values. The first hyperparameter is the number of trees, T . The other two hyperparameters penalize overly variable trees by applying some combination of the L1 penalty (α , analogous to LASSO) and the L2 penalty (λ , analogous to ridge regression). These can each be applied at full strength ($\alpha = 1$ or $\lambda = 1$), at

Table 1. XGBoost Hyperparameters

Parameter	Values
Number of trees (T)	1, 2, ..., 999, 1,000
L1 regularization coefficient on sample weights (α)	0, .5, 1
L2 regularization coefficient on sample weights (λ)	0, .5, 1

half strength ($\alpha = 0.5$ or $\lambda = 0.5$), not at all ($\alpha = 0$ or $\lambda = 0$), or some combination thereof.⁶

3.1.2. Random Forest

Random forests also use regression trees, but whereas XGBoost grows many trees sequentially and sums them, the random forest algorithm grows many trees independently and averages them (see James et al. 2013, chap. 8). We estimate 500 individual trees each using bootstrapped draws of training data. Random forests also decorrelates trees from each other by randomly sampling subsets of available covariates in each tree, ensuring that not all trees end up focusing on the same group of covariates. The predicted value for each observation is calculated separately for each of the 500 trees and averaged across trees to generate a single prediction for each test observation. We implement random forests using the ranger package in R.

3.1.3. LASSO

LASSO is a regularized version of linear regression. LASSO estimates regression coefficients by minimizing a loss function equal to the sum of squared prediction errors (like ordinary least squares, OLS) plus a penalty equal to a penalty parameter λ times the L1 norm of the standardized coefficient vector:

$$\sum_{i=1}^N (Y_i - \beta' x_i)^2 + \lambda \|\beta\|_1, \quad (2)$$

where all variables in the vector x_i have been standardized to have zero mean and unit variance, and $\|\beta\|_1$ is the L1 norm of the coefficient vector. The value of λ , known as the “LASSO hyperparameter,” is chosen to minimize out-of-sample prediction error as measured by the RMSE in the test sets. Notably the extreme case of $\lambda = 0$ corresponds to standard OLS, whereas a sufficiently high λ value will lead to β to be a vector of zeroes.

As is well known, by penalizing the L1 norm of the coefficient vector some (often many) coefficients can be driven exactly to zero, meaning the LASSO effectively performs

6. For more information on XGBoost parameters, see <https://xgboost.readthedocs.io/en/latest/parameter.html>.

some model selection by completely dropping variables to avoid overfitting. While XGBoost and random forests automatically capture interactions between explanatory variables, LASSO only considers linear functions of the input variables provided. Therefore, we form interactions manually by considering all possible one-way interactions between the variables discussed at the beginning of this section, which includes both weather variables and their squares, leading to a large set of 1,631 covariates for LASSO to consider, although many are dropped in CV.⁷ We implemented LASSO using the *glmnet* package in R.

3.1.4. RMSE Estimates

The RMSE estimates for each optimally tuned model are shown in table 2. The RMSE values pertain to the pretreatment period (June 2012–May 2013) as estimated using CV, meaning that they are indicative of each algorithm’s out-of-sample performance. XGBoost yields the lowest RMSE under all three samples, followed closely by random forests, and LASSO performs substantially worse. For this reason, we focus primarily on the results for XGBoost.

3.2. Generating Counterfactuals and Prediction Errors

After training and tuning each ML model on a year of pre-RCT data (i.e., June 2012–May 2013), we use each of the models to generate household-by-hour predictions for counterfactual energy consumption, denoted $\hat{Y}_{it}$ for each hour t in the experimental period (June 2013–October 2014). We then generate a “prediction error” for each such hour reflecting the difference between observed consumption and this counterfactual, or $Y_{it} - \hat{Y}_{it}$. We estimate the effect of treatment on this prediction error. Intuitively, if a treatment reduces observed consumption relative to the counterfactual, the prediction error will be negative, and the coefficient representing the treatment effect will reflect this negative effect. Alternatively, one could interpret the counterfactual prediction $\hat{Y}_{it}$ as a new pseudo-control observation in a TWFE or difference-in-differences framework.

To illustrate this design, figure 2 depicts the average hourly consumption profile on event days for households in the RCT control group (black solid line) and in the “treated” critical-peak pricing groups (gray solid line). We also plot the counterfactual prediction for those in the pricing group under the preferred ML approach, XGBoost, in the gray dashed line. The counterfactual prediction shown corresponds to the “treatment only” case where neither the control group nor the nonexperimental comparison group is used to generate the ML prediction, meaning the counterfactual is driven only by data from treated households. Our approach contends that the predicted counterfactual can generate a pattern that is comparable to the randomized control

7. Including all two-way interactions proved computationally infeasible, as it would result in nearly 24,000 regressors.

Table 2. Root Mean Square Error, by ML Approach and Sample (in kW)

	Randomized Control	Nonexperimental Comparison	Treatment Only
XGBoost	.6689	.7421	.6693
Random forests	.6825	.7592	.6972
LASSO	.8724	.9049	.8878

group. Visual inspection of figure 2 suggests that the counterfactual prediction is quite similar to the randomized control, particularly during the treated pricing period (shaded in gray). The divergence in energy consumption between the pricing and control group during the treated hours (4 p.m.–7 p.m., shown as the gray shaded area) is clear and presents visual evidence for a treatment effect. Absent the comparison to the control or comparison group, however, no strong, compelling evidence would exist for the existence or magnitude of the effect as demand during the peak hours, when peak period pricing is in effect, is actually higher than during the nonpeak hours. This situation is

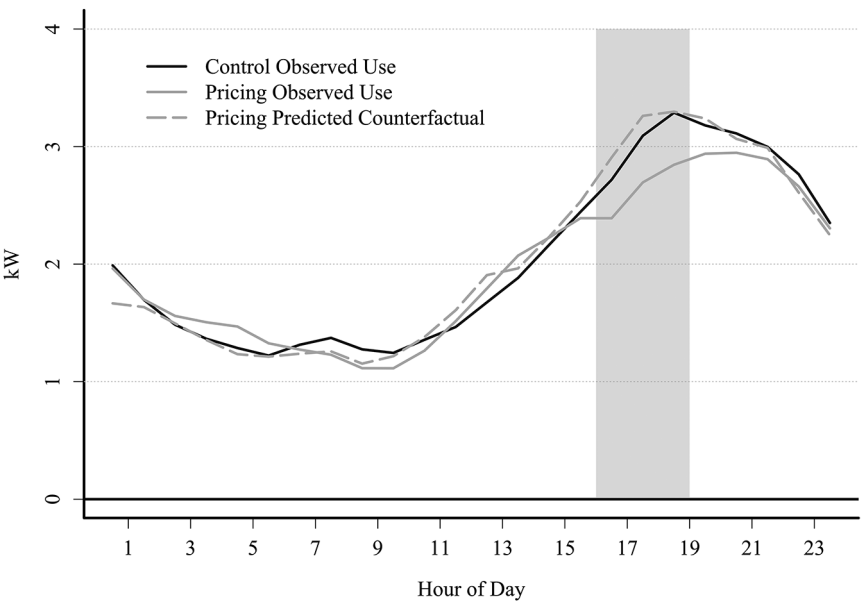


Figure 2. Event day average consumption profile for randomized control group, pricing group and XGBoost predicted counterfactual (using only data for treated households). The shaded region reflects peak hours (4 p.m.–7 p.m.). As in Burkhardt et al. (2019), the profiles are plotted after conditioning on household fixed effects and recentering on the grand mean to highlight the source of identification under a two-way fixed effects regression. Color version available as an online enhancement.

where the counterfactual predictions can substitute as a pseudo-control group. Comparing the observed consumption to its predicted counterfactual serves as evidence for the treatment effect, without requiring a randomized control group.

3.3. Treatment Effect Estimation

3.3.1. Two-Way Fixed Effects Specification

Once the prediction error is generated, we use standard two-way fixed effects (TWFE) approaches to estimate treatment effects, analogous to Burkhardt et al. but with the prediction error as our dependent variable. Namely, we regress the prediction error (observed consumption minus predicted) during the experimental period on indicators for treatment group, event day, and event hour indicators, and their interactions. We also include household fixed effects and flexible time fixed effects. Thus, we estimate the following equation:

$$Y_{it} - \hat{Y}_{it} = \sum_j \beta_j T_{ijt} + \mathbf{X}_{ijt} \gamma + \rho_i + \phi_t + \epsilon_{it}, \quad (3)$$

where Y_{it} and $\hat{Y}_{it}$ are actual and predicted consumption for household i at time t . Thus, $Y_{it} - \hat{Y}_{it}$ defines our prediction error, which serves as our outcome of interest. The treatment arm is denoted by j for the four arms (pricing, text message, text message + recommendation, and portal). The term T_{ijt} is an indicator variable equal to one for treated households in treatment arm j during peak hours (4 p.m.–7 p.m.) on treatment (“event”) days and zero otherwise. Hence, T_{ijt} can be thought of as the product of indicator variables for peak hour, treatment day, and treatment group, corresponding to a triple difference. The $\mathbf{X}_{ijt}$ term includes standalone peak hour and treatment day indicators, also interacted with treatment arm.

As is standard in TWFE settings, we also include unit- and time-specific fixed effects. The term ρ_i is a household fixed effect. In a prediction error setting, a household fixed effect captures the degree to which the ML model may over- or underpredict consumption for a specific household on average.⁸ The term ϕ_t represents time fixed effects, similarly representing the degree to which the ML model may over- or underpredict consumption across all households in a given period on average. As discussed in the next section, we consider cases with and without control or comparison data. In the former case, we use hour-of-sample fixed effects. Without a comparison group, we cannot use these fixed effects because they would be perfectly colinear with the treatment variable (i.e., 4 p.m.–7 p.m. on event days for treated households, which, without a comparison group, is all households). In this case, we use hour-of-day and week-of-sample fixed effects to capture diurnal and seasonal patterns. The results are not sensitive to alternative approaches to time fixed effects. We cluster standard errors at the household level.

8. As previously noted, we do not include household fixed effects in the prediction process because they lead to worse overall predictive performance than a “pooled” model that includes household characteristics did.

With a comparison group, this TWFE specification reflects a triple difference on the prediction error, analogous to Burkhardt et al.'s triple difference on electricity use, comparing treatment versus comparison group, treated day versus nontreated day, and treated hour versus nontreated hour. To see why this is a triple difference, we can walk through a very simple calculation on observed electricity use (as in Burkhardt et al.) that nonetheless approximately recovers the estimated pricing effect. For the treatment group, on event days, consumption averaged 1.7 kilowatts (kW) during non-peak hours but rose to 2.5 kW during peak hours, an increase of +0.8 kW from off-peak to peak. This is the first difference. The second difference is the analogous rise in consumption from off-peak to peak hours during nonevent days, also for the treatment group, which is a smaller +0.6 kW increase (1.8 kW vs. 1.2 kW). The difference in these differences is +0.2 kW, corresponding to a larger increase in consumption from off-peak to peak hours on event days among the treatment group. The third difference compares this increase for the treatment group relative to its no-treatment counterfactual; we can calculate the analogous value for the control group, for whom this relative off-peak to peak rise in consumption is +0.6 kW larger on event days (1.7 vs. 2.8 kW on event days, relative to 1.1 vs. 1.7 kW on nonevent days). The difference in these relative increases, $+0.2 \text{ kW} - 0.6 \text{ kW} = -0.4 \text{ kW}$, represents the third difference and is an approximation of the treatment effect estimated in equation (3).

Without a comparison group, the final difference (treated vs. comparison group) vanishes, meaning that the specification is a double difference. In both cases, the prediction error can be interpreted as an additional difference (observed consumption vs. predicted counterfactual). Intuitively, using the prediction error as the dependent variable tests not just whether observed electricity use, Y_{it} , was relatively reduced in the treatment period, but whether it was reduced relative to what one would otherwise predict from the ML prediction model based on historical data, $\hat{Y}_{it}$.

Finally, we also estimate regressions that do not use the predicted counterfactuals at all, in which case the left-hand side of equation (3) is simply observed consumption, Y_{it} , to show how standard, non-ML based methods fare under various conditions regarding the use of a comparison group, as discussed in the next section. Because equation (3) includes household and flexible time fixed effects on the right-hand side, both the standard approach (where the left-hand side is observed electricity use) and our ML-based prediction-error approach (where the left-hand side is the prediction error) can be interpreted as TWFE approaches. The only distinction is whether the dependent variable is observed electricity use or the ML-based prediction error. For brevity, we refer to the former as the standard TWFE approach, which routinely uses the outcome of interest as the dependent variable.

3.3.2. *Use of Control or Comparison Groups*

To explore when the prediction error approach is valuable for estimating treatment effects, we consider three different cases:

1. *Experimental control group*: We use data from households with pricing or information interventions and households that were in the experimental (randomized) control group (249 households total: 193 treated, 56 control).
2. *Nonexperimental comparison group*: We continue to use households that were treated, but we replace the experimental (randomized) control group with households not part of the experiment but in the same neighborhood, creating a nonexperimental comparison group (334 households total: 193 treated, 141 comparison).
3. *No control data*: We only use treated households. In this case, the predicted counterfactual consumption for each household-hour effectively serves as its own “control” (193 households total, all treated).

In each case, we retrain our three ML models using the pre-RCT data and generate prediction errors using only the relevant households. For example, this methodology ensures that the “no control” case corresponds to what would be practically implementable by a researcher who had access only to data on treated households and no control group, such as when a policy is simultaneously rolled out to an entire region or class of customers.

3.4. Criteria for a Successful Replication

Several criteria have been proposed to evaluate the quality of design-replication studies. Heckman et al. (1998) lay out four criteria of design that minimize bias from non-experimental studies: (H1) treatment and comparison units have the same distribution of unobserved attributes, (H2) treatment and comparison units have the same distribution of observed attributes, (H3) outcomes and characteristics are measured in the same way for both groups, and (H4) treatment and comparison units are drawn from a common economic environment. Cook et al. (2008) lay out several additional criteria for successful design-replication studies, which we adopt and interpret within the context of our nonexperimental framework. The primary Cook et al. (2008) criteria are (C1) a randomly chosen control group and a nonexperimental comparison group are included, (C2) the experiment and observational study should estimate the same causal quantity (e.g., an average treatment effect), (C3) the contrast between control and comparison groups should not be correlated with other variables related to the study outcome, (C4) analysts of the experimental and nonexperimental data should be blind to each other’s results, and (C5) and (C6), the experiment and nonexperimental analysis, respectively, should each meet standard criteria of technical adequacy.

In our context, the nonexperimental analysis meets all of the criteria of Heckman et al. (1998), except we have no way of knowing if we satisfy the first criterion, which is fundamentally untestable, although we control for time-invariant characteristics of the households with fixed effects to reduce the likelihood that household unobservables

confound our results (H1). Our treatment and comparison households possess similar levels and trends in pretreatment energy use (see, e.g., fig. 1) and our empirical design includes a variety of fixed effects as controls (H2); the data for all groups are generated from the same data-collection method (H3); and each of our groups is drawn from households in a similar neighborhood in Austin, TX, that all share a willingness to participate as Pecan Street households (H4). Our design also satisfies all but the fourth criterion of Cook et al. (2008). We construct a nonexperimental sample using Pecan Street households that were not subject to the experiment (C1) and our nonexperimental analysis estimates an average treatment effect, similar to the experiment (C2). The choice of experimental control and nonexperimental comparison sample is not correlated with the measured outcomes (C3). We do not satisfy C4, as we have knowledge of the experimental treatment effect estimates. The experiment was conducted soundly, and we found no reason to doubt the validity of its treatment effects (C5). Last, we conduct our nonexperimental analysis following common practices in the ML and program evaluation literature, documenting any design choices (e.g., choice of tuning parameters) and evaluating alternative specifications (C6).

The criteria described thus far focus primarily on study design, but they are not instructive for how to compare the resulting estimates from the nonexperimental analysis to the randomized benchmark. Cook et al. (2008) note that some judgment is required on how comparable the results of the experimental and observational analysis are. We adopt the following criterion explicated by Ferraro and Miranda (2017): the nonexperimental point estimate must fall within the 95% confidence interval of the experimental treatment effect.⁹ We also require that our nonexperimental estimates achieve the correct sign and significance as the randomized benchmark (providing correct inference for both nonzero and null effects), thus ensuring that these estimates additionally replicate the qualitative policy relevance (or lack thereof) of estimated treatment effects.¹⁰

9. We cannot generally conduct a formal test of the similarity of treatment effects between the experimental and nonexperimental estimates. This is because the two approaches are estimated using completely distinct methods, equations, and samples, which precludes a straightforward estimation of their joint uncertainties. If we were to ignore this covariance and simply use their individual standard errors, including that of the nonexperimental treatment effect, we might fail to reject equivalence between the two if the nonexperimental treatment effect were estimated with a large standard error. It would, in principle, be quite easy to find statistical equality between experimental and nonexperimental treatment effects with a very large standard error on the nonexperimental effect.

10. Ferraro and Miranda (2017) specify an additional criterion concerning the sensitivity of sample choice by incorporating a bootstrapping procedure to estimate nonexperimental treatment effects. Adopting this approach for our full sample would be computationally infeasible; nonetheless, we conduct a sensitivity analysis using subsampling as discussed in the next section.

One concern about using the randomized benchmark as “ground truth” is that treatment effects are stochastic, even in perfectly randomized settings. This makes the “ground truth” itself uncertain, meaning comparisons to it are imperfect. Therefore, we also conduct an alternative procedure based on an approach recommended by Smith and Todd (2005), who note that the treatment effect among untreated households is known to be zero. We therefore estimate the treatment effect on the control group using only the control group in the regression, in the same manner in which we estimate our nonexperimental treatment effects when using only treated households. This allows us to assess the ability of alternative nonexperimental methods (a control-only TWFE specification on consumption or on the prediction error) to recover a known null effect. This exercise could be thought of as a placebo test. We conduct this placebo test separately on both the randomized control group and the nonexperimental comparison group.

3.5. Testing Equality of Treatment Effect Estimates via Bootstrap

When constructing our prediction errors, there is additional error introduced because $\hat{Y}_{it}$ is estimated with error, which induces error in the treatment effects, β_j , not captured by conventional approaches to computing standard errors. We do not generally account for this additional source of error in our standard errors. There is no simple way to carry through the error introduced in the “first-stage” ML prediction process in our ad hoc approach. The same is true for any estimation that involves preprocessing an estimation sample (e.g., covariate matching algorithms paired with panel data estimation), although recent research provides some guidance (Abadie and Spiess 2022). Our primary approach relies solely on the standard error of the experimental benchmark to diagnose a successful replication (per Ferraro and Miranda 2017), and not the standard error of the ML-based prediction error approach, which mitigates this concern somewhat. This approach is, however, imperfect. Namely, the hypothesis we would ideally want to test is whether the treatment effect estimated using our ML-based prediction error approach using only treated households (denote this β_j^{ML} for treatment arm j) is the same as the treatment effect from the experimental benchmark RCT approach (β_j^{RCT}), or:

$$H_0: \beta_j^{\text{RCT}} = \beta_j^{\text{ML}}.$$

The test statistic that is strictly necessary to test this hypothesis depends not only on the variance of each treatment effect but also on their covariance. That covariance is particularly difficult to estimate because the two different treatment effect estimates are executed separately using different methods and data sets. In response to this issue, many researchers (including, e.g., Burlig et al. 2020; Christensen et al. 2021; and Abadie and Spiess 2022) rely on bootstrapping methods, sampling observations from a set of households, for example, in the first stage, re-running the ML prediction models and

the prediction error regressions, and then using the variation in estimated treatment effect coefficients to account for the variation introduced in the ML prediction. Unfortunately, conducting a bootstrapping exercise like this for our full sample is computationally onerous in our setting across all variants of our ML algorithms; with current estimation times, this process would take many months of computational time to complete with any reasonable number of bootstrap replications. For example, training just one of our ML routines (XGBoost) on the full sample to generate the results in table 3, including optimizing tuning parameters through cross validation, can take more than a day to run one iteration.

Therefore, as a sensitivity analysis, we conduct a more computationally feasible M-over-N bootstrap to test the null hypothesis (H_0 above) for the treatment-only case using our preferred ML procedure, XGBoost. Specifically, for each of 400 bootstrap

Table 3. XGBoost Prediction Error TWFE Regressions

	Experimental Control		Nonexperimental Comparison		Treatment Only	
	Use	Prediction Error	Use	Prediction Error	Use	Prediction Error
	(1)	(2)	(3)	(4)	(5)	(6)
Pricing	-.358***,†† (.091)	-.404***,††† (.092)	-.305***,†† (.080)	-.281***,† (.082)	.203** (.076)	-.361***,††† (.078)
Text Message + Recommendation	-.026 (.072)	-.092 (.073)	.027 (.058)	.040 (.060)	.540***,††† (.049)	-.048 (.055)
Text Message	.057 (.080)	-.083 (.078)	.111 (.068)	.016 (.066)	.618***,††† (.060)	-.048 (.061)
Portal	.023 (.075)	-.100 (.072)	.076 (.062)	.026 (.061)	.580***,††† (.054)	-.030 (.051)
Observations	2,627,869	2,627,869	2,996,735	2,996,735	2,056,604	2,056,604
Households	249	249	334	334	193	193

Note. Columns 1 and 2 are estimated using the original experimental sample. Columns 3 and 4 are estimated using the experimental treatment group and a nonexperimental comparison group. Columns 5 and 6 are estimated using the treatment group only. In the “Use” columns, the dependent variable is electricity use at the household-hour level. In the “Prediction Error” regressions, the dependent variable is the prediction error, also at the household-hour level. Standard errors are clustered at the household level. †’s indicate Bonferroni-adjusted significance for 36 tests.

† = $p \times 36 < .05$.

†† = $p \times 36 < .01$.

††† = $p \times 36 < .001$.

** = $p < .01$ under standard significance testing.

*** = $p < .001$ under standard significance testing.

iterations, we sample 25% of the households, with replacement.¹¹ For each iteration, we estimate the benchmark treatment effects from this subsample as in column 1 of table 3. Then, using only the treated households in the subsample, we retrain and optimally tune the XGBoost prediction algorithm, generate prediction errors, and compute treatment effect estimates as in column 6 of table 3. This results in 400 sets of paired treatment effect estimates (both the benchmark and the prediction-error-based replication using only treated households) for each of the four treatments. This allows us to formally evaluate the equality of the benchmark and prediction error estimates, accounting for both uncertainty in the ML prediction process and the covariance between the benchmark estimate and the replication.

4. RESULTS

We present our results in two formats. First, we present a table summarizing the detailed regression results for treatment effects from our preferred ML approach, XGBoost, which achieves the best predictive performance as measured by RMSE in CV.¹² After discussing the detailed results from this approach, we graphically summarize the results from the other ML approaches, which generally perform similarly. Detailed regression coefficients for each method are provided in the appendix (available online).

In table 3, we compare the results from the XGBoost prediction error regressions (in even-numbered columns) to those from the standard TWFE approach (in odd-numbered columns) applied to the electricity use data from the experiment. Column 1 shows the “ground-truth” estimates of the treatment effects of each intervention, reflecting the results from our standard TWFE regression using the original experimental sample (i.e., the original randomized treatment and control groups). Consistent with the results in Burkhardt et al., the pricing treatment produced a statistically significant reduction in peak consumption (about -0.36 kWh per hour, or -0.36 kW), but the other three treatments yielded no significant reduction.¹³ The asterisks in table 3 show

11. Since this 25% subsampling procedure yields samples that are each one-fourth the size of the original sample, the resulting variances are four times larger in expectation than those based on the full sample. Thus, appropriate inference, accounting for the true sample size, requires scaling down the resulting variance and covariance terms by a factor of four—or equivalently, scaling down the resulting standard errors by a factor of two.

12. XGBoost’s predictive accuracy is only slightly better than that of random forest. Both are superior to LASSO’s RMSE, but we include LASSO because it is common in the literature.

13. As noted, our sample is slightly different from the one used by Burkhardt et al. due to data updates by Pecan Street. In addition, to ease computational burden in training our ML models, we aggregate energy-use data to the hourly time step, whereas Burkhardt et al. used minute-level data. As a result of both of these differences, our treatment effect estimates are slightly different in magnitude from theirs (e.g., a pricing effect of -0.36 here vs. -0.39 in their analysis). Nevertheless, each treatment’s statistical significance (or lack thereof) is the same as in their study.

conventional statistical significance metrics, but because we are conducting many different hypothesis tests we also present Bonferroni-adjusted significance levels as dagger symbols.¹⁴ The results are robust to this correction.

Column 3 shows the results of an alternative regression that replaces the randomized experimental control with a nonexperimental comparison group. The results are similar to the experimental results in column 1, indicating that the nonexperimental control group serves as a good substitute in this setting. Column 5 presents estimated coefficients based on a regression that includes only data from treated households, where treatment effects are identified solely based on differences between treatment periods and nontreatment periods for treated households, conditional on household and time fixed effects.¹⁵ This column shows that when limiting the sample to just treated households, we do not recover the experimental results (i.e., col. 1); in fact, the results erroneously suggest that all the interventions increase energy use, because the interventions (such as critical-peak pricing) are triggered by periods of high demand. This structure means that without a comparison group, one would expect consumption for members of the treatment group to indeed be higher during periods of treatment than in other periods. The existence of a control or comparison group reveals that untreated households also have higher demand during these periods, allowing us to net out this effect. Without a comparison of some sort—whether an explicit control or comparison group or a predicted counterfactual—a comparison of treated and untreated periods would erroneously include this effect.

The results from the XGBoost prediction error regressions are displayed in columns 2, 4, and 6 of table 3 for the different samples. They indicate that this ML approach provides a good replication of the ground-truth results in all cases—a significant negative effect of the pricing intervention (at $p < .001$) and no effect of any of the information interventions ($p > .10$)—using both the experimental control and nonexperimental comparison groups. In all cases, the treatment effect estimates fall inside the 95% confidence intervals of the “ground-truth” results. Finally, with no comparison group, the XGBoost approach is able to replicate the experimental results using only data on treated households (col. 6), whereas the simple difference approach fails to do so (col. 5). Tables of treatment effect results for the other ML approaches are included in the appendix (tables A.1, A.2; tables A.1–A.11 are available online).

The bootstrap results shown in table A.3 demonstrate the robustness of our inference using the standard errors in table 3. In particular, those bootstrap results correspond

14. Specifically, in our prediction error approach, we estimate 36 treatment effects, reflecting four treatment effects, three ML algorithms, and three approaches to control groups. We therefore test these p -values with significance thresholds smaller by a factor of 36.

15. As previously discussed, all of these regressions include household fixed effects and time fixed effects (hour of sample in cases with a control group or hour of day and week of sample for regressions without a control group).

to formal tests of the equality of the estimates in columns 1 and 6 in table 3. In particular, column 3 of table A.3 shows that the prediction-error-based treatment effect estimates using only treated households are very similar to and statistically indistinguishable from the experimental benchmark estimates (the p -values for the null of no difference are .78, .98, .15, and .43 for the four respective treatments shown there). The bootstrapped standard errors on the differences are similar in magnitude to the standard errors on the individual point estimates, owing to a positive covariance between the benchmark estimates and their prediction error-based counterparts. This similarity lends credence to our main results in table 3, where we rely primarily on the confidence interval, and hence standard error, of the experimental benchmark to evaluate the success of a replication.

We show the results for XGBoost only in table 3, but we present results for other ML approaches in figures for each treatment to make comparisons across algorithms visually. Figure 3 shows the results for the estimated coefficient on the pricing treatment using our three ML approaches and for each of the three samples (with the experimental control group, the nonexperimental comparison group, and treatment group data only). The first black dot and line show the “ground-truth” treatment effect estimate and its 95% confidence interval, based on the standard TWFE regression on energy use (i.e., table 3, col. 1), to which all the other estimates should be compared. The figure reveals that all the ML approaches we use replicate the experimental findings for the pricing treatment and that the ML approaches do not perform substantially better in replication than using a nonexperimental comparison group in a standard program evaluation framework.

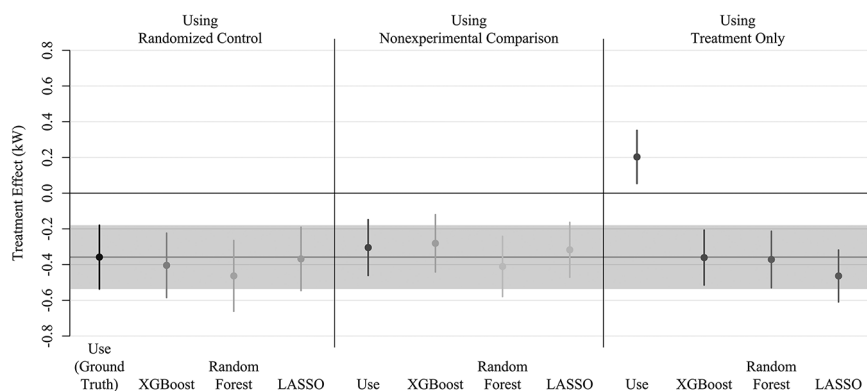


Figure 3. Critical-peak pricing treatment effect, by ML algorithm and sample. Labels along the bottom indicate the dependent variable used in the specification (eq. [1]). “Use” indicates that the dependent variable is electricity use. For the other cases, the dependent variable is the prediction error from the specified ML model. Point estimates and 95% confidence intervals are presented for each treatment effect estimate. The shaded area represents 95% confidence interval on ground-truth estimate. Color version available as an online enhancement.

While the ML approaches replicate the ground truth when the researcher has access to a reasonable, nonexperimental comparison group, they do not appear to add much value to the standard TWFE approach. By contrast, the right-most panel of figure 3 shows that ML approaches do add value in replicating experimental results when no comparison group is available. The simple differences regression erroneously suggests a significant positive effect of the pricing treatment on consumption (see table 3, col. 5), but all the ML approaches replicate the ground truth even without any control or comparison group. This finding suggests that ML approaches can be highly useful in situations where data are only available for treated observations. Intuitively, a counterfactual prediction can serve as a pseudo-control to replace control households.

Figure 4 provides analogous illustrations of the coefficient on the Portal treatment with and without comparison groups and shows similarly that (a) the TWFE approach with a nonexperimental control replicates the ground truth finding of no effect, (b) all the ML approaches with a control or nonexperimental comparison group replicate the ground truth, (c) the simple differences approach with no comparison group erroneously estimates a positive and significant treatment effect, and (d) all the ML approaches replicate the findings of no effect on consumption from access to the portal, as revealed by the original experiment. The corresponding figures for the remaining two treatment groups are similar and are shown in the appendix (figs. A.1, A.2; figs. A.1–A.4 are available online). Across the 36 treatment effect estimates (four treatments times three ML methods times three approaches to defining the control or comparison group), only two yield incorrect significance levels absent a Bonferroni adjustment. These two

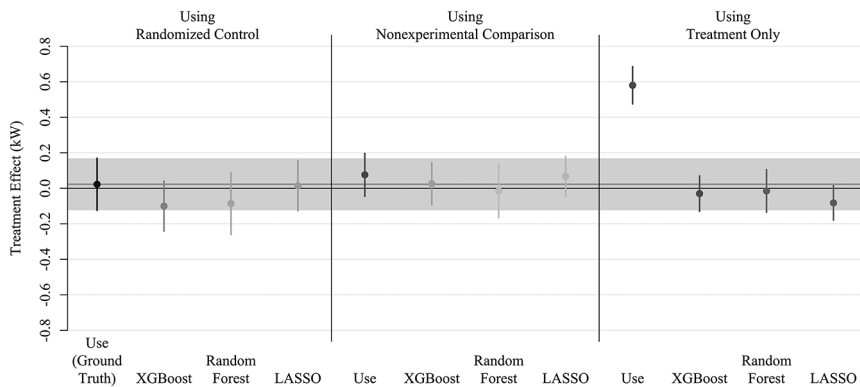


Figure 4. Portal treatment effect, by ML algorithm and sample. Labels along the bottom indicate the dependent variable used in the specification (eq. [1]). “Use” indicates that the dependent variable is electricity use. For the other cases, the dependent variable is the prediction error from the specified ML model. Point estimates and 95% confidence intervals are presented for each treatment effect estimate. The shaded area represents 95% confidence interval on ground-truth estimate. Color version available as an online enhancement.

both arise under the LASSO case (the worst-performing ML model) without a control or comparison group, where Text+Rec. and Portal treatments are both incorrectly found to be significant, at the 5% and 10% levels, respectively. At a 5% significance threshold, two false positives out of 36 is to be expected ($2/36 = 0.055$). Both of those false positives become insignificant when implementing a Bonferroni adjustment to the significance threshold, as would be expected.

As seen in column 5 of table 3, a naive regression using only treated households yields estimates of the wrong sign (positive) and significance. One reason for this is that the treatments, such as peak pricing or notifications, occur on particularly hot days when electricity demand is high due to air conditioning needs. The ML models account for this implicitly because they use data on temperature as an input and accordingly predict high counterfactual consumption.

There are two alternative, non-ML approaches that could help account for this positive treatment effect. The first would be to simply include flexible controls for temperature in the treatment effect estimation (eq. [3]). The results of this approach, which includes a quadratic temperature polynomial or flexible 5°F bins of temperature, are shown in table A.4. While this approach improves the estimates somewhat, it still generally fails to replicate experimental treatment effects. Even with flexible temperature controls, nearly all treatments, including those with true null effects, are incorrectly found to be significant at the 5% level or better. ML counterfactual prediction performs better across the board.

Another non-ML approach would be to continue to use our prediction error approach, but to simply generate predicted consumption using ordinary least squares regression, trained only on the pretreatment period, using the same set of variables considered by the other procedures (flexible time fixed effects, household characteristics, and quadratic polynomials of weather variables like temperature and humidity). The results of this approach are shown in table A.5. Again, the use of OLS-based prediction errors improves the estimates somewhat, but it again generally fails to replicate point estimates and yields incorrect inference, with two of the three treatments with null effects found to be statistically significant at the 5% level. ML is again found to be superior because it can capture unanticipated complex interactions between variables without overfitting.

Finally, we use the “treatment-only” specification on the control group to test the ability of each approach to recover known zero effects. As previously discussed, the “ground truth” estimates for the four treatment groups are estimated with error, making comparisons to them imperfect. By contrast, the treatment effect in the control group is known to be zero by construction. With this in mind, we use our “treatment-only” specification on the control group only, pretending they had been treated, effectively conducting a placebo test. The results are shown in table A.6, which shows that the non-ML approach applied only to the control group erroneously finds a large, positive treatment effect among the control group (0.56 kW, $p < .001$), while the

prediction error approach correctly yields a null result that is very close to zero (0.03 kW, $p = .55$). We also apply the same procedure to the nonexperimental comparison group, with similar results (see table A.7). This again illustrates the importance and value of the ML-based counterfactual prediction.

5. DISCUSSION

Our results provide strong evidence that ML prediction algorithms can replicate experimental benchmarks in several settings, for both null and nonzero treatment effects. In the most surprising case, we find that when we only use information on treated households, we can generate a predicted counterfactual that, when used in a prediction error regression, works as well as the randomized control group. This finding is powerful because all realms of economics have many instances in which policy changes occur for all units at the same time. Our finding suggests that ML counterfactual prediction algorithms provide some optimism for recovering causal effects of those policies even without an observational comparison group. Synthetic control methods are often used in scenarios like this, although our findings suggest that treatment effects can be recovered even without “donor” observations, and so it may be easier to meet the data requirements for implementing ML prediction algorithms.

In the case in which we have access to a group of nonexperimental units (i.e., units that were not included in the randomized experiment), we similarly find that we can replicate the experimental benchmarks using our ML-assisted prediction algorithms. Although this appears to be a valuable result, we find that a simpler, more standard TWFE design also recovers the experimental treatment effects. Therefore, if one has access to a reasonable comparison group, the ML-prediction-based methods do not improve upon simpler, more common, and easier-to-implement program evaluation tools. Fortunately, both ML and standard TWFE methods produce similar results, which would allow a researcher to be confident in their nonexperimental results if both frameworks were used. Our nonexperimental comparison group satisfies standard assumptions for TWFE/difference-in-differences (i.e., common pretrends), and our experimental treatment turns on at the same time for all units. If a researcher has a nonexperimental comparison group with different pretrends and poor balance among observables, then ML prediction methods (or other approaches to preprocessing a control group, such as covariate matching) may improve their ability to estimate causal effects.

Interestingly, because each of our ML algorithms generally replicates treatment effects in each scenario, we are unable to rank the algorithms’ ultimate effectiveness in estimating causal effects. We rely on standard CV and goodness of fit (i.e., RMSE) to provide guidance on choosing the algorithm with the best predictive qualities. Although we find that XGBoost performs better than random forest, which performs better than LASSO, all three reproduce the experimental benchmark, suggesting that marginal improvements in model fit are unlikely to be decisive for estimating causal effects.

Our findings lead to a fundamental question: in what settings are these results most likely to be applicable? Or, put another way, what features of a research design are most likely to benefit from ML counterfactual prediction algorithms? We explore these questions in our setting and speculate on the general features that will be most useful for researchers aiming to import our findings to other settings.

One aspect of this question revolves around what type of threats to identification the researcher is concerned about. In our setting, treatment is plausibly exogenous, conditional on household characteristics (or household fixed effects) and weather (or granular time fixed effects). The primary threat to exogeneity of treatment is that hot, summer days tend to be days when treatment kicks on to curtail consumption. Conditioning on weather or time fixed effects permits a plausible interpretation of exogeneity, and so our results can speak primarily to similar settings.

A common, alternative threat to identification is selection bias. It is important to highlight what kinds of selection bias our approach can and cannot mitigate. One type of selection concern is that households who sign up to participate in an experiment (whether they end up randomized into a treatment or control group) may be more likely to respond to the treatment, relative to the general population. This threatens the external validity of the findings, even if an internally valid treatment effect for recruited households can be reliably estimated. Our approach cannot solve this problem, which fundamentally stems from the design of a policy's recruitment process. Another type of selection bias occurs when households can select into treatment relative to non-treatment. This concern would bias comparisons between treated and untreated groups, threatening internal validity. The ML counterfactual approach can mitigate this problem because it allows for dropping the untreated group entirely, substituting ML counterfactual predictions.

Therefore, we anticipate that ML counterfactual approaches have the most promise to improve upon standard approaches when treatment is conditionally exogenous or selection into treatment is a concern, and in the latter case, the two methods are likely to produce different estimates. This distinction could explain why our panel and ML counterfactual results are comparable whereas, for example, the results from Burlig et al. (2020) for energy-efficiency upgrades in schools diverge between the two approaches.¹⁶ As a result, additional research in different settings that tests different

16. Burlig et al. (2020) study the energy savings from a Pacific Gas and Electric (PG&E) program that offered subsidies for energy efficiency retrofits of schools in California. This program was an opt-in program where school districts sought rebates to cover some of the costs of proposed retrofit measures. Issues of selection into treatment are a major consideration in using panel methods to evaluate the energy savings from this type of program. Indeed, we found that using a machine-learning approach to making counterfactual predictions produced a tighter range of estimated energy savings and smaller average savings from building upgrades than did the full range of panel models.

threats to internal validity is important to conduct before extrapolating our findings to settings in which selection into treatment is the primary concern.

The study of electricity demand has seen an increase in the application of ML algorithms primarily because (a) researchers can easily access high-frequency consumption data and (b) the data-generating process is predictable (e.g., with daily and seasonal patterns). Within this context, one question is how important granular data are (Ghanem and Smith 2021). Our experiment does not allow for answering this question, because the treatment period is a three-hour window during event days. Thus, we cannot aggregate our data, say, to the weekly or monthly level to evaluate how well the algorithms perform. The data can only be as aggregated up to the granularity of the treatment itself, so we must limit our focus to subdaily changes in energy consumption.

A second question is what type of data researchers need access to in order to generate causal estimates. How far would they get if they were limited in the number of explanatory variables to include in a study? We explore this question by evaluating variable importance plots from our XGBoost algorithm, presented in figure A.4. These plots iteratively exclude each variable (and hence its associated interactions) from the ML model and compute the resulting deterioration in predictive power. Across each sample, the most important variable is the square footage of the home, followed by the temperature, apparent temperature, capacity of solar PV installed on the home, and home age. The remaining top-10 characteristics include week or evening hour indicators and humidity. Notably, the structural characteristics of the home are time invariant and can be easily replaced by household fixed effects. Some combination of the weather variables is almost always included in electricity demand specifications based on theory and intuition, and these data are easily accessible from public databases or controlled for using temporal fixed effects. Recall that we replicated the experimental treatment effects in a standard TWFE design with fixed effects, which requires little information about household characteristics. As a result, it appears that little information about households is necessary for recovering treatment effects with nonexperimental data.

To explore this question a step further, we reestimate our primary XGBoost counterfactual prediction models several times, each time excluding certain categories of data. We then reestimate treatment effects using prediction errors from each restricted model. This is meant to emulate potential real-world data availability constraints that could be faced by researchers. These exclusions are summarized as follows:

1. Without household characteristic variables, using household fixed effects instead,
2. Without weather variables,
3. Without variables on weather or household characteristics, using household fixed effects (and time variables),
4. Using only time variables (time fixed effects and a weekly time trend).

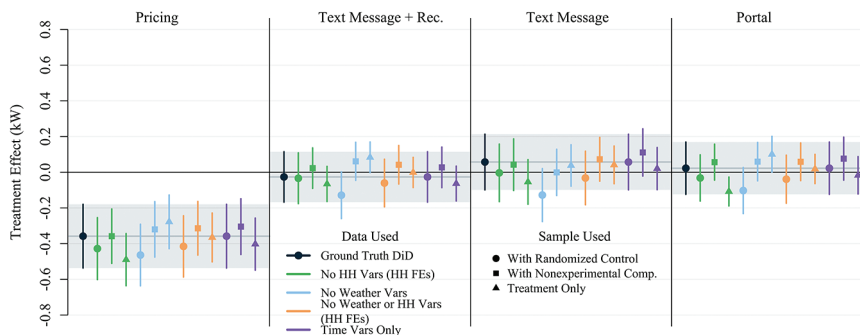


Figure 5. Prediction error-based treatment effect estimates with limited data in ML training. HH FEs = household fixed effects. Point estimates and 95% confidence intervals are presented for each treatment effect estimate. Shaded areas represent 95% confidence intervals on ground-truth estimate.

In each case, we rerun the prediction using the three samples in our main analysis: with the randomized control group, the nonexperimental comparison group, and treated households only. The results are shown in figure 5, alongside the experimental benchmark. Even with limited data, the ML prediction models perform well. Across 48 treatment effect estimates (four covariate constraints by three samples by four treatments), at the 5% significance level, we find no false negatives and only one false positive. At the 10% level, another four of the 48 estimates yield false positives, all of which appear when we drop weather data. The full regression results are shown in tables A.8–A.11. This set of results indicates that, in the context of electricity demand, the ML counterfactual predictions can do quite well even with little data. This result likely occurs because electricity demand shows strong diurnal and seasonal patterns, making it an attractive application for counterfactual prediction approaches. It is difficult to make strong claims about the full set of necessary conditions for ML counterfactual prediction to work well, other than to say that processes that can be predicted well with available data are good candidates for this approach.

6. CONCLUSION

The importance of sound program evaluation to policy makers and society at large cannot be overstated. Our work yields valuable policy insights and methodological contributions at the intersection of economics, energy policy, and ML that can inform that policy evaluation. While RCTs have played a growing role in policy evaluation, such approaches are not always feasible. At the same time, ML approaches are becoming more prevalent in such research. No prior formal comparisons of treatment effects estimated via counterfactual prediction methods that use ML and those estimated from randomized experiments exist. Our work begins to fill that void. The results of

our research help define best practices for evaluations of existing programs that may not have been implemented with randomized trials and can guide future analysis of consumer-focused energy policies.

When RCTs are not possible, identifying causal effects involves picking the right tools for the setting. Our results suggest that the contributions of ML methods to program evaluation and RCT replication depend importantly on the setting and on whether one has access to a suitable comparison group. When comparison group data are available, then familiar TWFE approaches perform comparably to an RCT, and thus may be a preferred approach. On the other hand, when data are only available for treated subjects, as may often be the case, then ML can provide a superior alternative to analysis based on simple differences in outcomes before and after treatment. Sometimes the right tool is a machine.

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